

Nathan Hershey

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Summary

Senior Full Stack Engineer with **8 years** of experience building and scaling distributed platforms across real estate, SaaS, and enterprise software. Led the design of cloud-native financial platforms processing high transaction volumes, implemented infrastructure automation with **Kubernetes** and **Docker** across multiple cloud providers, and built **CI/CD pipelines** reducing deployment time by 60%. Directed cross-functional teams of 5+ engineers while maintaining 99.9% uptime. Expert in **Python** and **Node.js** development with strong backend architecture, **AWS cloud infrastructure**, and real-time pipeline design for geospatial intelligence systems. Owns production infrastructure end to end, including on call coverage, deployment automation, and monitoring across distributed services. Comfortable partnering with platform and application teams on reliability work.

Core Skills

Languages and Frameworks: Python, JavaScript, TypeScript, Node.js, React, Next.js, RESTful APIs, GraphQL, HTML, CSS

Infrastructure and DevOps: AWS (EC2, S3, RDS, ECS, EKS, Lambda), Kubernetes, Docker

Databases and Messaging: PostgreSQL, MySQL, MongoDB, Redis, Kafka

Observability and Monitoring: Datadog, Prometheus, Grafana, CloudWatch, Sentry

Security and Architecture: OAuth2, JWT, SSL/TLS, distributed systems, microservices, event-driven architecture, air-gapped deployments, Bash, Linux, Terraform, Ceph

Work Experience

Senior Full Stack Engineer | Frontdoor | Real Estate Financial Platform | 09/2022 - Present

- Built and maintained CI/CD pipelines with **GitHub Actions** and **Terraform**, cutting deployment time by 60% and enabling zero-downtime releases for 40+ microservices.
- Established monitoring dashboards with **Datadog** and **CloudWatch**, achieving 99.9% platform uptime and reducing incident response time to under 5 minutes.
- Reduced page load latency from 4s to 1.5s by implementing **Next.js** performance optimizations and **AWS CloudFront** image CDN distribution.
- Designed and managed **Kubernetes** clusters on **AWS EKS** and **ECS**, orchestrating 15+ containerized microservices handling 200K+ daily requests.
- Architected event-driven microservices with **Python** and **Node.js** using **Kafka** message queues, improving system throughput by 40%.
- Led migration of monolithic applications to **AWS** cloud-native services, eliminating \$8K monthly in legacy infrastructure costs.
- Mentored 3 junior engineers on **Kubernetes** deployment patterns and **Docker** containerization best practices.

Full Stack Developer | ZenBusiness | SaaS Business Formation Platform | 01/2021 - 11/2022

- Developed 40+ **RESTful APIs** and **GraphQL** endpoints with **Node.js** and **React**, supporting 500K+ daily transactions across formation and compliance workflows.
- Optimized **PostgreSQL** query performance through indexing and query planning, improving report generation speed from 8s to 2s for 500K+ record datasets.
- Created Python automation scripts handling document generation and state filing workflows, saving 8+ engineering hours per week.
- Integrated third-party payment and compliance APIs with error handling and retry mechanisms, achieving 99.95% transaction success rate.
- Implemented JWT and **OAuth2** authentication systems protecting 5 enterprise partner integrations and 200K+ user accounts.

Full Stack Engineer | Northstar Software Development | Enterprise ATS Platform | 11/2020 - 06/2021

- Built applicant tracking system features with **Node.js** and **PostgreSQL**, serving 50+ recruiting agency clients and 10K+ monthly active users.
- Containerized and deployed applications on **AWS ECS** using **Docker**, reducing provisioning time from 2 hours to 15 minutes.
- Developed automated CI/CD pipelines with **GitHub Actions**, implementing automated testing that caught 95% of regressions before production.
- Optimized **AWS** infrastructure with **Terraform**, reducing monthly cloud costs by \$5K through right-sizing and resource tagging policies.
- Implemented **Redis** caching layer for job search queries, cutting average response time from 1.2s to 300ms.

Software Engineer | Arbisoft | Alert Notification Platform | 10/2018 - 08/2020

- Developed alert notification system features using **Node.js** and **Python**, serving 20+ enterprise clients in healthcare and logistics sectors.
- Designed **PostgreSQL** database schemas for 10+ client applications, implementing data models that scaled to 5M+ daily alert records.
- Created Python automation scripts for data import workflows, saving 5+ engineering hours weekly and eliminating manual processing errors.
- Built **RESTful APIs** with **OAuth2** and JWT authentication, reducing data breach risks through proper scope validation.
- Collaborated with cross-functional teams across 3 time zones to deliver FixAlert product features for 40+ projects.

Education

Bachelor of Science in Computer Science

University of North Texas | 2014 - 2018